Practical No. 2

Aim: Write a java program using conditional and looping statements.

Code:

Program 1:

```
import java.util.Scanner;

public class ConditionalsDemo {
   public static void main(String[] args) {
      Scanner scanner = new Scanner(System.in);

      // Input Year
      System.out.print("Enter a year: ");
      int year = scanner.nextInt();

      // 1. Using IF-ELSE to check for Leap Year
      if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
         System.out.println(year + " is a LEAP YEAR.");
      } else {
         System.out.println(year + " is NOT a leap year.");
      }

      // Input Month Number (1-12)
      System.out.print("Enter month number (1-12): ");
      int month = scanner.nextInt();

      // 2. Using SWITCH statement to determine the season
      String season;
      switch (month) {
         case 12: case 1: case 2:
            season = "Winter";
            break;
         case 3: case 4: case 5:
            season = "Spring";
            break;
         case 6: case 7: case 8:
            season = "Summer";
            break;
         case 9: case 10: case 11:
            season = "Autumn";
            break;
         default:
            season = "Invalid Month!";
            break;
      }

      System.out.println("Season: " + season);
      scanner.close();
   }
}
```

Program 2:

```
import java.util.Scanner;

public class LoopsDemo {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // 1. FOR LOOP: Print Fibonacci Series
        System.out.print("Enter number of terms for Fibonacci series: ");
        int terms = scanner.nextInt();

        int first = 0, second = 1;
        System.out.print("Fibonacci Series: ");

        for (int i = 1; i <= terms; i++) {
            System.out.print(first + " ");
            int next = first + second;
            first = second;
            second = next;
        }
        System.out.println("\n");

        // 2. WHILE LOOP: Check for Palindrome Number
        System.out.print("Enter a number to check if it's a Palindrome: ");
        int num = scanner.nextInt();

        int originalNum = num;
        int reversedNum = 0;



    }
}
```

Output:

Program1:

```
Enter a year: 2020
2020 is a LEAP YEAR.
Enter month number (1-12): 2
Season: Winter
```

Program2:

```
Enter number of terms for Fibonacci series: 4
Fibonacci Series: 0 1 1 2

Enter a number to check if it's a Palindrome: 123
123 is NOT a palindrome.
```